

```
PS C:\Users\user1\Desktop\dsa> ./blockchain.exe
```

```
=====
```

```
BLOCKCHAIN WITH PROOF OF STAKE
```

```
Currency: X Coin
```

```
=====
```

```
Warning: Invalid transaction not added to block
```

```
Genesis block created!
```

```
--- Step 1: Creating Wallets ---
```

```
Wallets created with initial balances.
```

```
===== ACCOUNT BALANCES =====
```

```
Address: Alice -> Balance: 1000 X
```

```
Address: Bob -> Balance: 800 X
```

```
Address: Charlie -> Balance: 1200 X
```

```
Address: Dave -> Balance: 500 X
```

```
=====
```

```
--- Step 2: Registering Validators ---
```

```
Validator Alice registered with stake: 500 X
```

```
Validator Bob registered with stake: 300 X
```

```
Validator Charlie registered with stake: 200 X
```

```
===== VALIDATORS =====
```

```
--- Validator Info ---
```

```
Address: Alice
```

```
Stake: 500 X
```

```
Total Rewards: 0 X
```

```
Status: Active
```

```
-----
```

```
--- Validator Info ---
```

```
Address: Bob
```

```
Stake: 300 X
```

--- Validator Info ---

Address: Bob
Stake: 300 X
Total Rewards: 0 X
Status: Active

--- Validator Info ---

Address: Charlie
Stake: 200 X
Total Rewards: 0 X
Status: Active

=====

===== ACCOUNT BALANCES =====

Address: Alice -> Balance: 500 X
Address: Bob -> Balance: 500 X
Address: Charlie -> Balance: 1000 X
Address: Dave -> Balance: 500 X
=====

--- Step 3: Creating Transactions ---

Transaction added to pending pool.
Transaction added to pending pool.
Transaction added to pending pool.
Transaction added to pending pool.

--- Step 4: Processing Transactions (Block 1) ---

Validator selected: Alice
Block #1 added to blockchain!
Reward of 10 X distributed to Alice

```
--- Step 4: Processing Transactions (Block 1) ---  
Validator selected: Alice  
Block #1 added to blockchain!  
Reward of 10 X distributed to Alice
```

```
===== ACCOUNT BALANCES =====  
Address: Alice -> Balance: 485 X  
Address: Bob -> Balance: 480 X  
Address: Charlie -> Balance: 975 X  
Address: Dave -> Balance: 570 X  
=====
```

```
--- Step 5: More Transactions ---  
Transaction added to pending pool.  
Transaction added to pending pool.
```

```
--- Step 6: Processing Transactions (Block 2) ---  
Validator selected: Bob  
Block #2 added to blockchain!  
Reward of 10 X distributed to Bob
```

```
===== ACCOUNT BALANCES =====  
Address: Alice -> Balance: 365 X  
Address: Bob -> Balance: 610 X  
Address: Charlie -> Balance: 895 X  
Address: Dave -> Balance: 650 X  
=====
```

```
--- Step 7: Even More Transactions ---  
Transaction added to pending pool.  
Transaction added to pending pool.  
Transaction added to pending pool.
```

```
--- Step 8: Processing Transactions (Block 3) ---
```

```
--- Step 7: Even More Transactions ---
Transaction added to pending pool.
Transaction added to pending pool.
Transaction added to pending pool.

--- Step 8: Processing Transactions (Block 3) ---
Validator selected: Alice
Block #3 added to blockchain!
Reward of 10 X distributed to Alice

===== ACCOUNT BALANCES =====
Address: Alice -> Balance: 345 X
Address: Bob -> Balance: 550 X
Address: Charlie -> Balance: 1025 X
Address: Dave -> Balance: 610 X
=====

--- Step 9: Validating Blockchain Integrity ---
Blockchain is VALID!

--- Step 10: Display Complete Blockchain ---

=====
                BLOCKCHAIN STATE
=====
Total Blocks: 4
Total Validators: 3
Total Stake: 1000 X
=====

===== BLOCK 0 =====
Hash: b6b493c7cc7a07c122154148da8d1c8f8d63e04c1f730e837d51c2df3d849ff6
Previous Hash: 0
Validator: GENESIS
Timestamp: 1763667155
```

===== BLOCK 0 =====

Hash: b6b493c7cc7a07c122154148da8d1c8f8d63e04c1f730e837d51c2df3d849ff6

Previous Hash: 0

Validator: GENESIS

Timestamp: 1763667155

Transactions: 0

Transactions in this block:

=====

===== BLOCK 1 =====

Hash: edade9c73cd39601dc3b929f1795b7adf482b87e2371af2e3af6d4849cc58b41

Previous Hash: b6b493c7cc7a07c122154148da8d1c8f8d63e04c1f730e837d51c2df3d849ff6

Validator: Alice

Timestamp: 1763667155

Transactions: 5

Transactions in this block:

Transaction ID: 6cac1e2339c9fa9b4559fed35a57cadd1d3d2d91f2d4f4f44298872bdbad8a25

From: Alice

To: Dave

Amount: 100 X

Timestamp: 1763667155

Signature: dfff377603a9a242...

Transaction ID: 3be634176b2a3836e7ff2e2e1e3f201d13b651c2e633db286378600870c1e1ac

From: Bob

To: Charlie

Amount: 50 X

Timestamp: 1763667155

Signature: 96d04fe6518e5ebd...

Transaction ID: 8670d6dd836a4dfaa7f2e176484a78bce91f6e7c4dd4dcea6f2fa0771dede4e3

From: Charlie

A
G

Transaction ID: 8670d6dd836a4dfaa7f2e176484a78bce91f6e7c4dd4dcea6f2fa0771dede4e3
From: Charlie
To: Alice
Amount: 75 X
Timestamp: 1763667155
Signature: c1e7c2afa19ac19a...

Transaction ID: 250ef1d500948a7544231ce8a678cf36607e773d896441869eadf4357577516b
From: Dave
To: Bob
Amount: 30 X
Timestamp: 1763667155
Signature: fa0313697e31402f...

Transaction ID: 7e2cbd5df13926066d025b4164455fae3ef0543e09981866e9cf57c11663b0db
From: SYSTEM
To: Alice
Amount: 10 X
Timestamp: 1763667155
Signature: 2506de77bb666580...

=====

===== BLOCK 2 =====

Hash: dc8a96acdf2ab831a5a5a1aeaa45835060335fadcd2fb7d0cf4704ff79efcaa3
Previous Hash: edade9c73cd39601dc3b929f1795b7adf482b87e2371af2e3af6d4849cc58b41
Validator: Bob
Timestamp: 1763667155
Transactions: 3

Transactions in this block:

Transaction ID: 0d5f2e9f12c3ae9bb44824a02b410c9bc2d2e91d30a9802f49ce1a474e6ca33c
From: Alice
To: Bob
Amount: 120 X
Timestamp: 1763667155

Run and Debug (Ctrl+Shift+D)

--- Final Validator Stats ---

===== VALIDATORS =====

--- Validator Info ---

Address: Alice

Stake: 520 X

Total Rewards: 20 X

Status: Active

--- Validator Info ---

Address: Bob

Stake: 310 X

Total Rewards: 10 X

Status: Active

--- Validator Info ---

Address: Charlie

Stake: 200 X

Total Rewards: 0 X

Status: Active

=====

--- Final Account Balances ---

--- Validator Info ---

Address: Charlie

Stake: 200 X

Total Rewards: 0 X

Status: Active

=====

--- Final Account Balances ---

===== ACCOUNT BALANCES =====

Address: Alice -> Balance: 345 X

Address: Bob -> Balance: 550 X

Address: Charlie -> Balance: 1025 X

Address: Dave -> Balance: 610 X

=====

=====

BLOCKCHAIN DEMO COMPLETED

Total Blocks: 4

=====

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